The rise and roots of music streaming

Music streaming is huge today with dozens of services vying for your attention, but the very concept is much older than you'd have thought. In this new section dubbed "origins" we trace the roots and sometimes tumultuous stories behind today's everyday technologies.

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In 1897, a certain Thaddeus Cahill made a device called the Telharmonium – a device that took up a massive room with 200 tons of machinery and wires. It served as a centre for broadcasting music via telephone wires. Initially called the Dynamophone, it worked as an on-demand music broadcasting service where subscribers had to call and request for the Telharmonium line, following which they would be linked to the station. The Telharmonium would play electrically generated tunes over the line, which would be amplified at the user's end with a paper funnel.

This was the first true instance of music streaming, even before the invention of the very first amplifier in 1906. The Telharmonium generated live music at a 'music plant' at Broadway and 39th, and operated across the city of Manhattan. The music was eerie, and yet, attained a point of novelty and bewilderment among the crowd who noticed it. In December 1906, the New York Times wrote, "It is scientifically perfect music, capable of reproducing any sound produced by any musical instrument and many more that no musical instrument produces. Sounds like the extravagant exaggeration of a charlatan, doesn't it?"

Despite the first working model of the magnetic tape, the Telegraphone, being developed in 1898, there are no known recordings, live demos or any kind of content that accurately replicates how the Telharmonium may have sounded like. Arthur T. Cahill, brother of Telharmonium's inventor Thaddeus Cahill, was reportedly looking for a place to house the mammoth prototype instrument in the 1950s. Unfortunately, he could not find a place, and the Telharmonium disappeared into obscurity. It left its legacy behind with the Hammond Organ, designed in the 1930s, which took its working principle from the Telharmonium.

Following the acceptance of Cahill's Telharmonium, the jukebox and the radio were the next steps in the history of music streaming. The very first jukebox can be traced back to even before the telharmonium, and was the first instance of a device transmitting recorded music. The jukeboxes used a coin-operated mechanism to produce music recorded in metal discs and cylinders. Take for instance the nickel-in-slot Edison Class M Electric phonograph invented by Louis Glass and William Arnold – inserting a coin and turning a lever would set a turntable of sorts in motion and place the stylus pin at the first groove of the disc or cylinder, following which music could be heard via listening tubes (headphones of the present day).


The legendary Telharmonium that streamed music into your telephone receiver

Over time, by the 1920s, jukeboxes started incorporating automation. In 1927, the Automated Musical Instrument company, later known as AMI, introduced a selective music-playing jukebox, which could hold multiple records and change tracks automatically. Known as phonographs till then, the term "jukebox" only came into being in the 1940s in the United States of America, with the rise of electrical automation, recording and amplification. Popular song listings came into being, in light of the most-frequently-played tracks on these jukeboxes. Wallboxes were yet another important feature of jukeboxes, giving remote control-like features to a listener. Towards the 1960s, these wallboxes started incorporating speakers with stereo sound technology.

Radio transmission developed parallelly, with broadcasts incorporating requests for recorded audio. Much like music streaming services of today, radio channels and broadcasters began striking deals with music labels to gain rights for playing their music. The introduction of radio services in the 1920s saw a drastic paradigm shift for music labels in terms of profitability of music. With more listeners being allured by the prospect of free music, radio stations began figuring out ways to earn profits by introducing advertisements. In the meantime, vinyl record sales for music labels fell from a valuation of $75 million in 1929 to as low as $5 million in 1933, aided by the Great Depression. The music labels started filing lawsuits against radio stations, citing the reason that the free music was drawing listeners to the stations, and putting ads before playing music led to commercial gain from the copyrighted tracks. The copyright holders had a right over profitable public performances of tracks, and stations were enforced to pay